

EXP7: Breadth First Search (BFS)

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Aim:

To develop a Python program that implements the Breadth First Search (BFS) algorithm to traverse a graph level by level starting from a given source node.

Algorithm:

Step 1: Represent the graph using an adjacency list (dictionary of lists).

Step 2: Create an empty queue and insert the starting node into it. Mark it as visited.

Step 3: Remove the front node from the queue and add it to the traversal result.

Step 4: Visit all unvisited neighbours of the removed node, mark them as visited, and insert them into the queue.

Step 5: Repeat Steps 3 and 4 until the queue becomes empty.

Step 6: Display the BFS traversal order.

Source Code:

```
# Program to implement Breadth First Search (BFS)
```

```
from collections import deque
```

```
def bfs(graph, start):
    visited = set([start])
    queue = deque([start])
    order = []

    while queue:
        node = queue.popleft()
        order.append(node)

        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

    return order
```

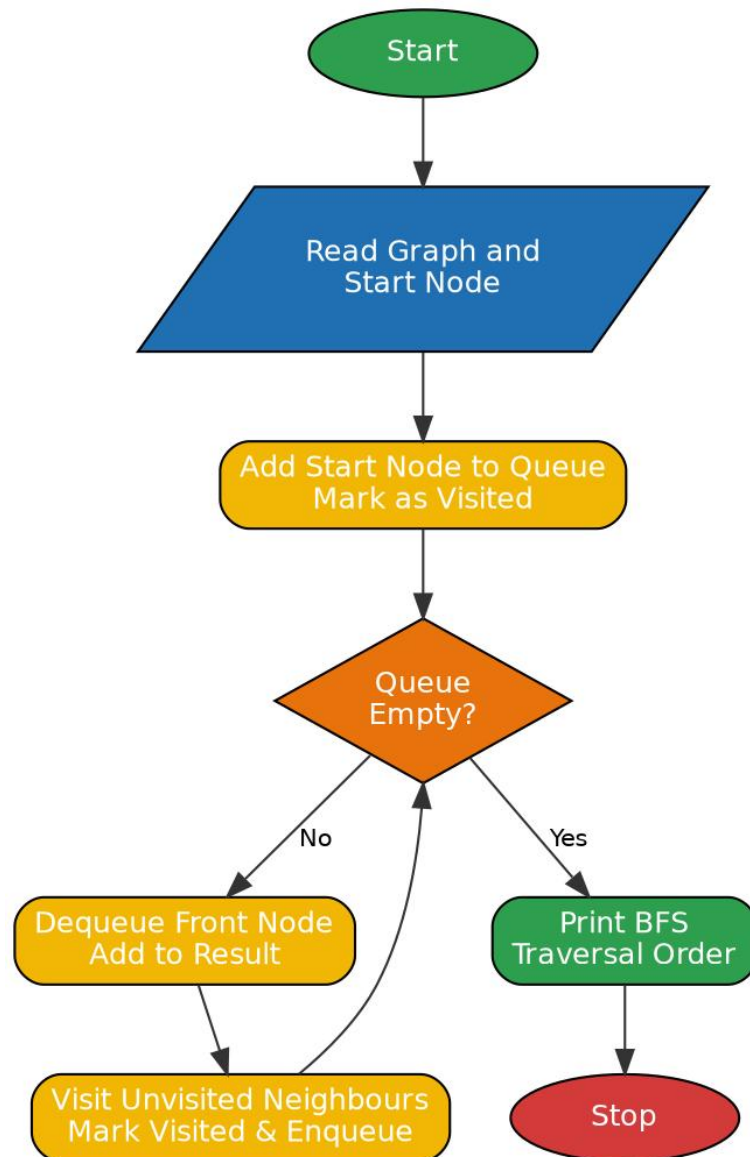
```
graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'F'],
    'D': ['B'],
    'E': ['B', 'F'],
    'F': ['C', 'E']
}
```

```

start_node = 'A'
result = bfs(graph, start_node)
print("BFS Traversal starting from", start_node, ":")
print("-> ".join(result))

```

Flowchart:



Output:

```

>>> ===== RESTART: C:/Users/yanar/Documents/CSA1717/EXP7.py =====
BFS Traversal starting from A :
A -> B -> C -> D -> E -> F

```

Result:

Thus the program to implement Breadth First Search (BFS) was written, executed successfully and the output was verified.